
Your answers should be i) precise, ii) to-the-point, and iii) with legible handwriting.

1. Suppose a quadcopter seeks a landing zone which is marked with many bright lamps. The quadcopter has a brightness sensor with the following details:

- Binary sensor measurement: $z \in \{\text{bright}, \neg\text{bright}\}$
- Sensor model:

$$p(z = \text{bright} \mid x = \text{home}) = 0.6$$

$$p(z = \text{bright} \mid x = \neg\text{home}) = 0.3$$

- Binary state: $x \in \{\text{home}, \neg\text{home}\}$
- Prior: $p(x = \text{home}) = 0.5$

Assume that the robot observes *light*. What is the probability of the robot is above its home ? In the next time step, robot observes $z = \neg\text{bright}$. What is the probability of robot is above its home ?

2+2

2. Suppose a mobile robot is moving on a long straight road. Our location x will simply be the position along this road. Suppose that, initially, the robot is believed to be at location $x_{\text{init}} = 1,000m$, but we happen to know that this estimate is uncertain. Based on the uncertainty, we model our initial belief by a Gaussian with variance $\sigma_{\text{init}}^2 = 900m^2$.

To find out more about our location, we query a GPS receiver. The GPS tells us our location is $z_{\text{GPS}} = 1,100m$. The GPS receiver is known to have an error variance of $\sigma_{\text{GPS}}^2 = 100m^2$.

- i) Write the pdfs of the prior $p(x)$ and the measurement $p(z \mid x)$. 2
- ii) What is the posterior $p(x \mid z)$? 2

3. Write the *Markov localization* algorithm. 3

4. Consider a simplified global localization in the following grid-style environments (Figure 1). In each environment the robot will be placed at the random position facing *North*. Come up with an open-loop localization strategy that contains a sequence of the following commands:

Action *L*: Turn left 90 degrees

Action *R*: Turn right 90 degrees

Action *M*: Move forward until you hit an obstacle

At the end of this strategy, the robot must be at a predictable location. For each such environment, provide the *shortest* such sequence (only “M” actions count). State where the robot will be after executing your action sequence, for the two environments in Figure 1. You can number the grid cells for your convenience. 3

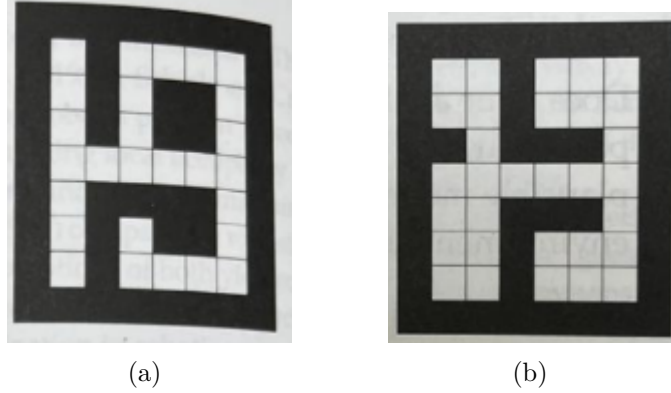


Figure 1: Grid-style environments

5. In *Monte Carlo localization* the expected value of the location computed by the algorithm differs from the true expected value.

Consider a world with four possible robot locations: $X = \{x_1, x_2, x_3, x_4\}$. Initially, we draw $N = 5$ samples uniformly from among those locations. Let Z be a Boolean sensor variable with the following conditional probabilities:

$$p(z \mid x_1) = 0.8, \quad p(z \mid x_2) = 0.4, \quad p(z \mid x_3) = 0.1 \text{ and } p(z \mid x_4) = 0.1.$$

Assume that we only generate one new sample in the resampling process, regardless of N . This sample might correspond to any of the four locations in X . Thus, the sampling process defines a probability distribution over X .

What is the resulting probability distribution over X for this new sample? 3

6. In *Occupancy Grid mapping*, the idea is to calculate the posterior over possible maps given the data. Consider a discrete workspace with fine-grained grids. Each grid cell has a binary occupancy value. Write down the necessary equations for estimating the occupancy information using *log odds ratio* for a *static* workspace. 3